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Total Productive Maintenance (TPM) Implementation in a Machine Shop: A Case Study

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Abstract

Quality and Maintenance of manufacturing systems are closely related functions of any organization. Over a period of time two concepts have emerged which are Total Productive Maintenance (TPM) and Total Quality Management (TQM) along with other concepts to achieve World Class Manufacturing system. In this paper experience of implementing Total Productive Maintenance is shared and investigated for a company manufacturing automotive component. Concept is implemented in the machine shop having CNC turning centers of different capacity. Overall Equipment Effectiveness is used as the measure of success of TPM implementation. The losses associated with equipment effectiveness are identified. All the pillars of TPM are implemented in a phased manner eliminating the losses and thus improving the utilization of CNC machines.

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Keywords: Total Productive Maintenance, Total Quality Mamagement, 5-S, Kaizen.

Nomenclature

A	Availability
OEE	Overall Equipment Effectiveness
PE	Performance Efficiency
QR	Quality Rate

1. Introduction

In today's industrial scenario huge losses/wastage occur in the manufacturing shop floor. This waste is due to operators, maintenance personal, process, tooling problems and non-availability of components in time etc. Other forms of waste includes idle machines, idle manpower, break down machine, rejected parts etc. are all examples of waste. The quality related waste are of significant importance as they matter the company in terms of time, material and the hard earned reputation of the company. There are also other invisible wastes like operating the machines below the rated speed, start up loss, break down of the machines and bottle necks in process. Zero oriented concepts such as zero tolerance for waste, defects, break down and zero accidents are becoming a pre-requisite in the manufacturing and assembly industry. In this situation, a revolutionary concept of Total Productive Maintenance (TPM) has been adopted in many industries across the world to address the above said problem.

The goal of the any TPM program is to improve productivity and quality along with increased employee morale and job satisfaction. Earlier preventive maintenance was considered as non value adding process, but now it is essential requirement for longer life cycle of machines in an industry. TPM is an innovative approach to maintenance that optimizes equipment effectiveness, eliminates breakdowns, and promotes autonomous operator maintenance through day-to-day activities involving the total workforce.

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Nakajima S.[1] done pioneering work and has given basic definition of TPM, its importance, goals of TPM, objectives of TPM, merits and demerits of TPM and steps to be followed while implementing TPM. Also author has described about challenging limits for TPM, method for calculation of OEE, possible areas of wastage of resources which may occur. I. P. S. Ahuja et al.[2] gives in-depth review on TPM literature published. Author has summarized eight pillars for the success of TPM implementation as shown in Fig. 1. F. Ireland et al.[3] given a study of total productive maintenance implementation in three companies. Nakajima's seven steps of autonomous maintenance was the focus for implementation. Ross Kennedy et al.[4] give the concept of TPM^3 means third generation TPM which is having eight pillars instead of seven pillars. TPM^3 is an enhanced Australian approach applying the principles and practices of the Toyota Production System and the Toyota Way - Lean and TPM. Ki-Young jeong et al.[5] gives the detailed loss classification scheme for computing OEE for a capital intensive industry. F. T. S. Chan et al.[6] implemented the concept of TPM in semiconductor industry. Authors have reported 83% improvement in equipment productivity. Marcelo Rodrigues et al.[7] discussed the reasons of failure of TPM implementation and concluded that the shallow involvement of the people of various levels is the main reason of the failure of TPM. G. Chand et al.[8] reported to implement the concept in cellular manufacturing system having forming shop, tool room and product test room. In the paper they have reported 62% OEE and concluded the need of sustained TPM implementation to achieve world class OEE level of greater than 85%. M. C. Eti et al.[9] implemented the concept in Nigerian industry and concluded that TPM can succeed only if necessary training is provided and time is given to monitor the success or failure of improvement activities.

In this paper experience of implementing TPM concept in automotive manufacturing company (RBD Engineers Ltd., Khatraj, Gandhinagar, Gujarat) is discussed. All the pillars of TPM are implemented in a phased manner leading to continuous improvement of the company. In the section 2 various pillars are explained and approach taken to implementation is also mentioned. Section 3 discusses the analysis of the data before and after implementation of the TPM. Section 4 gives the conclusion of the present work.

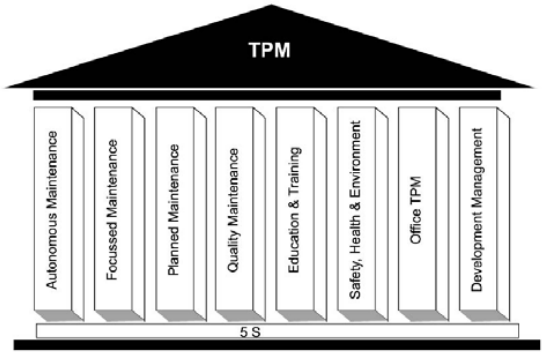


Fig. 1 Pillars of TPM [2]

2. TPM Implementation

TPM concept is implemented in a phased manner in a machine shop of a company manufacturing automotive components. In each phase one TPM pillar is implemented. Overall equipment effectiveness (OEE) is taken as a measure of success of TPM implementation. Approach of implementation of each pillar is discussed in following subsections.

2.1. 5-S

TPM starts with 5S. 5S can be called as foundation stone of TPM implementation. It is a Japanese way of housekeeping. Problems cannot be recognized is the work place is unorganized. Cleaning and organizing the workplace helps us to pop up the problems. Making problems visible and seen to the people gives an opportunity of improvement. If this 5S is not taken up seriously, then it leads to 5D i.e. Delays, Defects, Dissatisfied customers, Declining profits and Demoralized employees. Meaning of each 'S' is explained in Table 1.

Table 1: Meaning of 5-S

Japanese Term	English translation	Equivalent 'S' term
Seiri	Organization	Sort
Seiton	Tidiness	Systematize
Seiso	Cleaning	Sweep
Seiketsu	Standardization	Standardize
Shitsuke	Discipline	Self-discipline

Approach:
The concept of 5S is introduced to the Quality Circle formed in the company and to the workers as well. Current situation of the workplace was studied by 5S audit using check sheets and ratings were calculated. Table 2 gives the detailed 5S implementation report.

Table 2: 5S Implementation Report

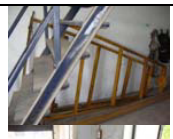
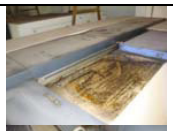
Report	Before		After	
1S Report	Crane used to load chuck for bell housing is still kept in between CNC machines even after its use is over.		Crane is taken to the corner of the shop floor after its use is over thus eliminating the interference during normal operation.	
	Coolant tank which was delivered by CNC manufacturer by mistake was lying in machine shop.		Tank is stored in central store room. The space is free for storing accepted parts.	
	Chip conveyor tank which was delivered by CNC manufacturer by mistake and lying in machine shop.		Tank is stored in central store room. The space is free for storing defective parts.	
	Waste bins, plastic bag and rejected parts are placed in corner near the staircase.		The place is utilized by Brinnel Hardness testing machine for inspection purpose.	
	Waste boxes, bins, chair and plastic bag is kept behind CNC- 01.		That much area is used to keep accepted and rejected parts.	
	Unwanted angles and pipes are kept near the window of machine shop.		That much area is used to keep accepted and rejected parts.	
	Rejected parts are kept besides machine.		The parts are removed and the space is freed.	-
2S Report	Earlier patches on the floor were disturbing material movement using trolley.		Patches are filled with Cement thus helping smooth material flow.	
	Trolleys are randomly placed anywhere.		Space is dedicated for keeping trolleys whenever not needed.	
	No labeling is done and tools are placed randomly in racks.		Labeling is done and tools are stored in their respective places identified.	
	Allen keys, nut bolts and other tools are kept randomly.		Proper places are decided for each tool.	
	No red tag marking is done for rejected materials.		Red tag marking is done and wall is also white washed.	
	Plastic bags and waste cotton is randomly kept.		They are kept systematically.	

Table 2: 5S Implementation Report (contd.)

Report	Before		After	
3S Report	CNCs are not cleaned from top side.		Cleaning CNCs from top also is included in Daily Maintenance.	
	Coolant overflows from the tank.		It is cleaned and overflow problem is solved.	
	Operators used to spit in corner of machine shop after chewing tobacco.		Dust bin is now relocated and wall is white washed.	
4S Report	No operator is writing hourly report.	-	Writing hourly report is compulsory.	
	Employee details are not displayed on the notice board.	-	Employee details are displayed on the notice board	
	No working information is displayed on the notice board.	-	Working instructions, control process plan daily maintenance sheet and part drawing are displayed on each CNCs.	
	Jaws and blocks were kept randomly.		Jaws and blocks of every CNCs are kept in the standard format.	
	There was no standard cycle time displayed in CNC shop.	-	The standard cycle time of the job is prepared.	
5S Report	-	-	Company's Mission and Vision statements are displayed in Hindi as well as Gujarati.	
	-	-	Suggestion scheme stating that whoever gives the best suggestion will be given reward of Rs 500/-.	

2.2. JISHU HOZEN (Autonomous Maintenance)

This pillar is based on the concept that if operators take care of small maintenance tasks it will free up the skilled maintenance people to concentrate on more value added activity and technical repairs. The operators are responsible to upkeep their equipment on daily basis to prevent it from deteriorating. By use of this pillar, the aim is to maintain the machine in new condition. The activities involved are simple nature like cleaning, lubricating, visual inspection, tightening of loosened bolts etc.

Objective of JISHU HOZEN (Autonomous maintenance) are operation of equipments without breakdown, versatile and flexible operators to operate and maintain other equipments and eliminating the defects at source through active employee participation.

Approach:

- Operators are given training by shop floor in charge in a pair of two about daily maintenance.
- Earlier helper use to clean the machines, and they are not cleaning it regularly, as a result of which lots of work is accumulated during preventive maintenance.

- Operator and helper are cleaning machine from its top and now they are responsible to upkeep the machine on which they are working.
- Main key for autonomous maintenance is operator's mentality; he will take good care of machine only if he thinks as if he owes that machine.
- With the help of maintenance department, shop floor in-charge and manuals of CNCs, Autonomous Maintenance Check list is prepared. Now following this checklist is mandatory for all the operators. It is a responsibility of shop floor in-charge to see that every operator fills up this check list.

2.3. Planned Maintenance

It is aimed to have trouble free machines and equipments without any breakdown and producing components to the quality level giving total customer satisfaction. Maintenance can be carried out as Preventive Maintenance, Breakdown Maintenance, Corrective Maintenance and Maintenance Prevention. Planned Maintenance is a proactive approach which uses trained maintenance staff to help train the operators to better maintain their equipment. Objective of Planned Maintenance are to achieve and sustain availability of machines, optimum maintenance cost, improve reliability and maintainability of machines, zero equipment failure and break down and ensure availability of spares all the time.
Approach:

- Any major breakdown of mechanical or electrical component is responsibility of CNC manufacturer. So preventive maintenance and predictive maintenance is being carried out in house.
- For preventive maintenance all the helpers of CNC shop floor and maintenance department along with engineers of maintenance department are given training every week i.e. on Thursday when preventive maintenance is scheduled, by Head of department of Maintenance department and Shop floor in charge.
- Predictive maintenance is done in collaboration of CNCs operators and engineers of Maintenance department.
- Operators are given training to identify the areas in CNC which are more likely to fail, so that they can inform Maintenance department in advance and take corrective action before any major break down.

2.4. Kaizen

"Kaizen" literally means "change for the betterment". Kaizen involves small improvements and is carried out on a continual basis and involving people of all level in the organization. The principle behind Kaizen is that "a very large number of small improvements are more effective in an organizational environment than a few improvements of large value". This pillar is aimed at reducing losses in the workplace that affect our efficiencies. By using a detailed and thorough procedure we eliminate losses in a systematic method using various Kaizen tools. Objective of Kaizen is achieve and sustain zero loses with respect to minor stops, measurement and adjustments, defects and unavoidable downtime.
Approach:

- Poka Yoke device:- It is Japanese term in English which means mistake proofing or error prevention. Poka Yoke device can be of two types - Warning / prevention type and detecting type. Tool board is made in-house as a Poka Yoke device as shown in Figure 2. It is warning type mistake proofing devices which will visually warn operator or helper if they are try to put spanners at the wrong place.



Figure 2: Poka Yoke

- Coolant leakage problem:- To identify the reasons of coolant leakage fish bone diagram is prepared as shown in Fig 3. Corrective action to solve above identified problems is given in Table 3.

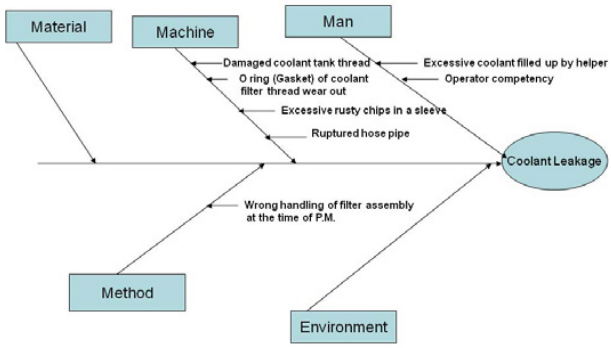


Figure 3: Fish bone diagram

Table 3: Analysis for fish bone diagram

Problem	Occurrence	Risk	Severity	Corrective Action
Coolant leakage from filter	1. Due to O-ring (gasket) in the filter head got damaged 2. While tightening filter back into filter head is filter is not held at 90° and because of aluminum casted body and cross taper its threads are damaged.	Moderate	High	New O-ring has been ordered. Proper training is given to helpers who are doing weekly preventive maintenance.
Coolant leakage from tank	1. One of the punching nut of coolant tank got damaged may be due to application of over pressure on the nut while tightening it. 2. Overflow from tank due to overfilling of coolant by helper.	Moderate	High	Gasket in the tank is changed and new punching nut is installed. Training and number of bucket to be filled is clearly specified to the helper
Coolant leakage from hose pipe	Due to it's to and fro motion along with turret and it has certain life span after which it gets ruptured.	Low	Low	Installing new hose pipe

- New Layout: Management has decided to purchase two new CNCs to meet the production requirement. As the company is planning of two new machines, it is necessary to redesign the layout of CNC shop floor. Present and proposed layouts are shown in Fig 4 and 5 respectively. Proposed layout is designed to minimize the handling of parts.



Figure 4: Present layout

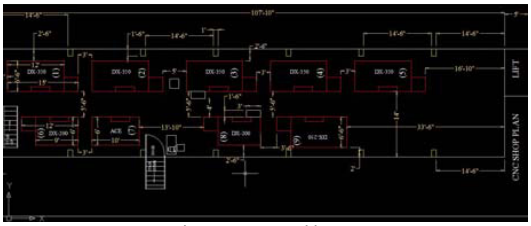


Figure 5: Proposed layout

2.5. Quality Maintenance

It is geared towards achieving customer satisfaction through delivery of highest quality product. Through focused improvement defects are eliminated from the process after identifying the parameter of machine which affects the product quality. Transition is from Quality Control to Quality Assurance.

Approach:

The condition is checked and measured in time series to verify that measured values are within standard values to prevent defects. The transition of measured values is watched through charts to predict possibilities of defects occurring and to take counter measures before hand.

2.6. Training

Approach:
Continuous improvement is possible only through continuous improvement in knowledge and skill of the people as different levels. To reduce the defects training is provided to the concerned as show in Table 4.

Table 4: Training

To Operators (CNC shop floor)	To Shift incharge (Quality Department)
Training given for 100% visual inspection to detect visual defects Training given for using "Go and No go" gauges for 100% inspection and also for educating them how to use measuring instrument for e.g. vernier caliper, screw gauge and dial gauge etc. Training given by shop floor incharge regarding necessary force required to clamp the job into the chuck. Training given optimum set up for machine including how to mount chuck on the machine, in which order reduction sleeves, blocks and jaws should be installed etc. They are also advised to do setting at mean value given in the control copy of the drawing, earlier they setting machine at minimum value or we can say lower control limit, and thus causes rejection	Weekly training for quality department regarding QC tools and its implementation and also for educating them how to use measuring instrument.

2.7. Office TPM

Office TPM is the pillar which follows other four other pillars of TPM (JH, Kaizen, QM, PM). Office TPM must be followed to improve productivity and efficiency of the administrative functions. This includes analyzing processes and procedures which can be automated. Office TPM addresses nine major losses which are processing loss, cost loss including in areas such as procurement, accounts, marketing, sales leading to high inventories, communication loss, idle loss, set-up loss, accuracy loss, office equipment breakdown, communication channel breakdown, telephone and fax lines and time spent on retrieval of information.
Approach:

- Now operators are writing daily rejected quantities in their daily report, so that time required in getting data for daily rejection from Quality department is reduced.
- New computer system is proposed for Maintenance department because Maintenance department is writing its daily report in a notebook and then that notebook is given to administration and then they document it and also for making schedules for maintenance they have to wait for person in administration to document it and take its hard copy. And also their chance of loss of old records having valuable data.
- Earlier daily insert usage report was maintained in a notebook, and at the end of the month the person has to sit along with calculator to sum up the monthly consumption. Now the report is prepared in MS Excel saving lots of time and effort.
- Now employee details are displayed on the notice board having information like their name, designation, phone number so that whenever anybody need to consult them can reach them on their mobiles, without wasting time.

2.8. Safety, Health and Environment

The purpose of this pillar is to create a safe workplace and a surrounding area that is not damaged by our process or procedures. Utmost importance to Safety is given in the plant. Manager - safety department looking after functions related to safety. Objective of this pillar are to achieve zero accident, zero health damage and zero fires.
Approach:

- Sufficient number of fire extinguisher is provided all over CNC shop floor.
- Training is given to each and every individual about how to use fire extinguisher in case of emergency in every 6 months.
- Management is given suggestion for giving training to employee what to do in case of emergency? What should be the exit plan?
- Management is also given suggestion to conduct mock drill once in a year.
- Earlier water camper in the shop floor was washed weekly now helpers are advised to wash it within 2 days.
- Workers are advised to maintain cleanliness of toilets, regular cleaning of toilets is also done.
- Workers are also advised not to chew tobacco and spit it in CNC shop floor and not to smoke within company premises.

3. Data Analysis

Overall Equipment Effectiveness (OEE) is widely use as the measure of success of TPM implementation. Overall Equipment Effectiveness is given as: Availability × Performance Efficiency × Quality Rate ($OEE = A \times PE \times QR$)[1]. Availability takes into account the losses due to equipment failure and setup and adjustment and is calculated as the ratio of operating time to loading time. Performance Efficiency includes losses due idling and minor stoppages and speed loss and is calculated as ratio of net operating time to operating time. Quality rate factors the defects in process and reduced yield and is defined as ratio of valuable operating time to net operating time. Table 5 gives the summary of observation made before and after TPM implementation.

Table 5: Summary of observation

Sr. No.	Category	Before implementation	After implementation
1	Shift time	720 min.	720 min.
2	Total production in a shift	160 Nos.	72 Nos.
3	Scheduled break	50 min.	50 min.
4	Non-scheduled break	5 min.	0 min.
5	Breakdown	4 min.	0 min.
6	Cleaning, inspecting & tightening of insert	15 min.	15 min.
7	Operator absent	55 min.	6.5 min.
8	Non-conforming product	5 Nos.	3 Nos.
9	Theoretical cycle time	3 min.	7.5 min.
10	Availability (A)	0.90	0.98
11	Performance Efficiency (PE)	0.73	0.85
12	Quality Rate (QR)	0.96	0.95
13	OEE ($A \times PE \times QR$)	0.63	0.79

4. Conclusion

Following conclusion is derived from implementation of TPM in the machine shop of automotive company:

1. Success of TPM depends on various pillars like 5-S, Jishu Hozen, Planned Maintenance, Quality maintenance, Kaizen, Office TPM and Safety, Health & Environment.
2. Overall Equipment Effectiveness has improved from 63% to 79% indicating the improvement in productivity and improvement in quality of product.
3. It is observed that most of the defective components are because of the previous process namely casting hence to improve the productivity efforts must also be given to previous process as well.
4. The key factors for this implementation are workers involvement and top management support. Still world class TPM implementation is possible with continual support at all the levels along with the supply of necessary resources.

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